

September 2023 Exam

EC400

Probability and Statistical Inference

Suitable for all candidates

Instructions to candidates

This paper contains SEVEN questions. Answer ALL questions.

At the top of your answer scripts, please write the following:

- Your full name;
- Your class group; and
- Your class teacher's name.

Time Allowed: *2 hours*

You are supplied with: *Standard Normal and Student's t tables*

Calculators: *Calculators are NOT allowed in this examination*

1. **Total: 15 points (18 minutes)**

Let $\{A_1, A_2, \dots\}$ denote a set of events defined on sample space Ω . Let $\Pr(A_i)$ denote the probability of event A_i , and let A_i^c denote the complement of A_i . Let $\emptyset$ denote an empty set. Use the notation given above to answer the following questions.

(a) State the axioms of probability. **(3pts)**

(b) Prove that for any two events A_1 and A_2 ,

$$\Pr(A_1 \cup A_2) = \Pr(A_1) + \Pr(A_2) - \Pr(A_1 \cap A_2).$$

(6pts)

(c) Now suppose we toss a fair coin twice. Let us define three events as follows:

A_1 : the first toss is heads,

A_2 : the second toss is heads, and

A_3 : the first and second tosses both show the same side of the coin.

Show that events A_1, A_2 and A_3 are pairwise independent.

(4pts)

(d) Are events A_1, A_2 and A_3 (as defined in part (c)) fully independent? **(2pts)**

The answer for part (a) is:

Axiom 1. $\Pr(A_i) \geq 0$ for $i = 1, 2, \dots$; **1 point**.

Axiom 2. $\Pr(\Omega) = 1$; and **1 point**.

Axiom 3. For any events A_i and A_j , where $i \neq j$, such that $A_i \cap A_j = \emptyset$, it holds that

$$\Pr\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \Pr(A_i).$$

1 point.

The answer for part (b) is:

We have $A_1 = (A_1 \cap A_2) \cup (A_1 \cap A_2^c)$, and $A_2 = (A_1 \cap A_2) \cup (A_1^c \cap A_2)$ and $(A_1 \cup A_2) = (A_1 \cap A_2^c) \cup (A_2 \cap A_1^c) \cup (A_1 \cap A_2)$. In other words, we have expressed A_1, A_2 and their union as unions of mutually disjoint events. **1 point for each of the three equations**.

It is then immediate with repeated use of Axiom 3 that

$$\begin{aligned} \Pr(A_1 \cup A_2) &= \Pr(A_1) - \Pr(A_1 \cap A_2) \\ &\quad + \Pr(A_2) - \Pr(A_1 \cap A_2) \\ &\quad + \Pr(A_1 \cap A_2), \end{aligned}$$

and the required equality follows. 3 points for this or equivalent derivations.

The answer for part (c) is:

A_1 and A_2 are clearly independent events. We can also see that $\Pr(A_3) = \Pr(\{HH, TT\}) = 1/2$ and $\Pr(A_1 \cap A_3) = \Pr(\{HH\}) = 1/4 = \Pr(A_1)\Pr(A_3)$, so A_1 and A_3 are independent. Similarly, A_2 and A_3 are independent. We deduce that $\{A_1, A_2, A_3\}$ are pairwise independent.

2 points for details of A_1 (or A_2) versus A_3 and 1 point for a similar argument for A_2 (or A_1) versus A_3 . 1 point for mentioning A_1 versus A_2 explicitly even if it is trivial.

The answer for part (d) is:

To show mutual independence, we need to check two conditions:

- Any subset of two or more events are (mutually) independent.
- $P(A_1 \cap A_2 \cap A_3) = P(A_1)P(A_2)P(A_3)$

We have already proved that the first condition holds (it's just pairwise independence), but the second condition does not hold:

$$P(A_1 \cap A_2 \cap A_3) = P(\{HH\}) = 1/4, \text{ but } P(A_1)P(A_2)P(A_3) = (1/2)^3 = 1/8.$$

We deduce that the events are not mutually independent. 2 points for derivations in previous line and conclusion.

2. Total: 15 points (18 minutes)

(a) Consider the density function for random variable X given by

$$f_X(x) = \begin{cases} \alpha k^\alpha / x^{\alpha+1}, & \text{for } x \geq k, \\ 0, & \text{otherwise,} \end{cases}$$

where $\alpha > 1$ is an unknown parameter and $k > 0$ (the smallest possible value of X) is a known constant.

Derive the cumulative distribution function, $F_X(x)$. **(7pts)**

(b) Consider a different density function for random variable X given by

$$f_X(x) = \frac{1}{2} \exp(-|x|), \text{ for } -\infty < x < \infty.$$

Show that the moment-generating function of X is given by

$$M_X(t) = \frac{1}{(1-t^2)},$$

and state the values of t for which $M_X(t)$ is well-defined. **(8pts)**

(a) We have:

$$\begin{aligned} \int_{-\infty}^x f(t) dt &= \int_k^x \frac{\alpha k^\alpha}{t^{\alpha+1}} dt = (-k^\alpha) \int_k^x (-\alpha) t^{-\alpha-1} dt \\ &= (-k^\alpha) [t^{-\alpha}]_k^x \\ &= (-k^\alpha) (x^{-\alpha} - k^{-\alpha}) \\ &= 1 - k^\alpha x^{-\alpha} \\ &= 1 - \left(\frac{k}{x}\right)^\alpha. \end{aligned}$$

Therefore:

$$F_X(x) = \begin{cases} 0 & \text{for } x < k \\ 1 - (k/x)^\alpha & \text{for } x \geq k. \end{cases}$$

Award 2 points for correct initial limits of integration and setting up the cdf formula with an appropriate change of variable - e.g. “dt” instead of “dx”. Award 3 points for above (or equivalent) derivations. Award 2 points for formal and full statement of the CDF at the end (note: if a genuine attempt has been made to formally specify the CDF in full – i.e. over all possible values of x – then award these last two points irrespective of whether the student has made an error in the derivations).

(b) We have:

$$\begin{aligned} M_X(t) = E(e^{tX}) &= \int_{-\infty}^{\infty} e^{tx} f(x) dx = \int_{-\infty}^{\infty} e^{tx} \frac{1}{2} e^{-|x|} dx \\ &= \int_{-\infty}^0 \frac{1}{2} e^{(t+1)x} dx + \int_0^{\infty} \frac{1}{2} e^{(t-1)x} dx \\ &= \frac{1}{2(t+1)} [e^{(t+1)x}]_{-\infty}^0 + \frac{1}{2(t-1)} [e^{(t-1)x}]_0^{\infty} \\ &= \frac{-2}{2(t+1)(t-1)} \\ &= \frac{1}{1-t^2} \end{aligned}$$

provided $|t| < 1$.

Award 2 points for evidence of knowledge of correct formula for MGF.

Award 2 points for considering positive and negative sections of support separately.

Award 2 points for final derivations.

Award 2 points for correctly stating values of t for which the MGF is well-defined.

3. Total: 10 points (12 minutes)

Consider the bivariate density function

$$f(x, y) = \begin{cases} \frac{3}{2}, & x^2 \leq y < 1, 0 \leq x \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

- (a) Find the conditional density function of Y given $X = x$. **(5pts)**
- (b) Find $\Pr(X \geq 1/2, Y \geq 1/2)$. (You may leave your final answer in terms of simplified fractions.) **(5pts)**
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For part (a):

We need to use the result that $f(y | x) = \frac{f(x, y)}{f(x)}$. Obtain $f(x) = \int_{x^2}^1 (3/2) dy = \frac{3}{2}(1 - x^2)$ with $0 \leq x \leq 1, 0$ otherwise. Therefore $f(y | x) = \frac{1}{1 - x^2}$ for $x^2 \leq y < 1$ and $0 \leq x \leq 1$. Award 1 point for evidence of knowledge of a conditional pdf - even if general formula is not explicitly stated. Award 2 points for marginal pdf. Award 2 points for final answer.

For part (b):

$$\begin{aligned} & \Pr(X \geq 1/2, Y \geq 1/2) \\ &= \Pr(X \geq 1/2, Y \geq 1/4) - \Pr(X \geq 1/2, 1/4 \leq Y < 1/2) \\ &= \int_{1/2}^1 \int_{x^2}^1 3/2 dy dx - \int_{1/2}^{1/\sqrt{2}} \int_{x^2}^{1/2} 3/2 dy dx \\ &= \int_{1/2}^1 \left(\frac{3}{2}(1 - x^2) \right) dx - \int_{1/2}^{1/\sqrt{2}} \left(\frac{3}{2} \left(\frac{1}{2} - x^2 \right) \right) dx = \frac{5}{8} - \frac{1}{4}\sqrt{2}. \end{aligned}$$

Award up to 3 points for correct approach and/or correct limits of integration and/or

the sketch of the region over which integration is required is correct. Award the final 2 points for students who arrive successfully at the final answer.

4. **Total: 15 points (18 minutes)**

- (a) Consider an estimator $\hat{\mu}$ for unknown parameter μ . Define formally the statistical properties of bias, variance and mean-square error of $\hat{\mu}$. Derive an exact mathematical relationship between these three properties. **(5pts)**
- (b) Now suppose $\hat{\mu}_1$ and $\hat{\mu}_2$ are unbiased estimators of μ such that $\text{Var}(\hat{\mu}_1) = \sigma_1^2$ and $\text{Var}(\hat{\mu}_2) = \sigma_2^2$. A third estimator, $\hat{\mu}_3$, is proposed. It is defined as

$$\hat{\mu}_3 = k\hat{\mu}_1 + (1 - k)\hat{\mu}_2$$

for some constant k such that $0 < k < 1$.

- i. Evaluate the bias, variance and mean-square error of $\hat{\mu}_3$. **(3pts)**
- ii. Suppose that $\hat{\mu}_1$ and $\hat{\mu}_2$ are independent. For what choice of k is $\hat{\mu}_3$ optimal in a minimum mean-square error sense? (You do **not** need to verify the second-order condition.) **(4pts)**
- iii. Provide intuition for the expression you obtained in part ii. above. **(3pts)**

The answer to part (a) is:

$$\text{Bias}(\hat{\mu}) = E(\hat{\mu}) - \mu$$

$$\text{Var}(\hat{\mu}) = E((\hat{\mu} - E(\hat{\mu}))^2)$$

$$\text{MSE}(\hat{\mu}) = E((\hat{\mu} - \mu)^2)$$

1 point for each of the three definitions. so that, using the binomial theorem and linearity of expectations, we have

$$\begin{aligned}\text{MSE}(\hat{\mu}) &= E(((\hat{\mu} - E(\hat{\mu})) + (E(\hat{\mu}) - \mu))^2) \\ &= \text{Var}(\hat{\mu}) + \text{Bias}^2(\hat{\mu}) + 2E((\hat{\mu} - E(\hat{\mu}))(E(\hat{\mu}) - \mu)) \\ &= \text{Var}(\hat{\mu}) + \text{Bias}^2(\hat{\mu}).\end{aligned}$$

2 points for proof.

For part (b)i, we have:

$$\begin{aligned} E(\hat{\mu}_3) &= E(k\hat{\mu}_1 + (1-k)\hat{\mu}_2) = kE(\hat{\mu}_1) + (1-k)E(\hat{\mu}_2) \\ &= k\mu + (1-k)\mu \\ &= \mu \end{aligned}$$

hence bias of $\hat{\mu}_3$ with respect to μ is zero. 1 point. Further, we have, due to independence:

$$\begin{aligned} \text{Var}(\hat{\mu}_3) &= k^2 \text{Var}(\hat{\mu}_1) + (1-k)^2 \text{Var}(\hat{\mu}_2) \\ &= k^2 \sigma_1^2 + (1-k)^2 \sigma_2^2. \end{aligned}$$

1 point. Since $\hat{\mu}_3$ is unbiased, its mean-square error is equal to its variance. 1 point.

For part (b)ii, minimising $\text{Var}(\hat{\mu}_3)$ is the same as minimising the mean squared error. Differentiating, we obtain as our first-order condition:

$$2k^* \sigma_1^2 - 2(1-k^*) \sigma_2^2 = 0$$

2 points. which yields

$$k^* = \frac{\sigma_2^2}{\sigma_1^2 + \sigma_2^2}$$

2 points. as the optimal solution.

For part (b)iii, we observe that the third estimator is a convex combination of the first and second estimators, where the weight assigned to the first estimator increases as the relative variance of the second estimator increases. In other words, we are combining two unbiased estimators in the most efficient way possible by up-weighting the relatively precise estimator and down-weighting the relatively noisy estimator. Therein lies the source of optimality in the MMSE sense. 3 points for any explanation that adequately captures the gist of this answer.

5. **Total: 10 points (12 minutes)**

Suppose we have a set of mutually independent random variables $Y_i \sim \text{Normal}(i\mu, i^3\sigma^2)$ for $i = 1, \dots, N$, where μ is unknown but σ^2 is known. Derive the maximum likelihood estimator for μ . (You do **not** need to verify the second-order condition.) (10pts)

Hint: You may find it helpful to recall that if $X \sim \text{Normal}(\mu, \sigma^2)$, then

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right) \text{ for any } x \in \mathbb{R}.$$

2 points for likelihood; 2 for log-likelihood; 2 for first derivative; 2 for final answer; 2 for precision and clarity of overall logic/argumentation (e.g. deduct one of these two marks if they don't put a hat on μ when setting score function to zero and/or deduct a mark if their equations are hard-to-follow).

$$\begin{aligned}\mathcal{L} &= \prod_{i=1}^N \frac{1}{\sqrt{2\pi(i^3\sigma^2)}} \exp\left(-\frac{(y_i - i\mu)^2}{2i^3\sigma^2}\right) \\ &= (2\pi)^{-N/2} \sigma^{-N} \left(\prod_{i=1}^N i^{-3/2}\right) \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^N i^{-3}(y_i - i\mu)^2\right), \text{ so} \\ \log \mathcal{L} &= -\frac{N}{2} \log 2\pi - N \log \sigma - \frac{3}{2} \sum_{i=1}^N \log i - \frac{1}{2\sigma^2} \sum_{i=1}^N i^{-3}(y_i - i\mu)^2, \text{ thus,} \\ \frac{\partial \log \mathcal{L}}{\partial \mu} &= -\frac{1}{\sigma^2} \sum_{i=1}^N i^{-3}(y_i - i\mu)(-i).\end{aligned}$$

Setting the score to zero implies that

$$\begin{aligned}\hat{\mu}_{ML} \sum_{i=1}^N i^{-1} &= \sum_{i=1}^N i^{-2} Y_i, \text{ so that our ML estimator is} \\ \hat{\mu}_{ML} &= \frac{\sum_{i=1}^N i^{-2} Y_i}{\sum_{i=1}^N i^{-1}}.\end{aligned}$$

6. Total: 20 points (24 minutes)

Let X denote the lifespan (in years) of a given mouse. Assume that X is distributed Normal(μ, σ^2). Suppose σ^2 is unknown but it is known that $\mu = 3.4$ years. A scientist, Sara, claims that the mean lifespan of African pygmy mice is greater than 3.4 years. She plans to test her claim using a sample of 9 African pygmy mice.

- (a) Define H_0 and H_1 . Suggest a test statistic and state its distribution under H_0 . Are any additional assumptions needed? **(5pts)**
- (b) Sara obtains sample average, $\bar{x} = 3.7$, and sample variance, $s^2 = 1/9$.
 - i. Given a 5% level of significance, discuss whether Sara can reject the null or not. Interpret the finding in your own words. **(5pts)**

- ii. What is the p -value of the test? Give your best approximation. **(5pts)**
- iii. If the true mean lifespan of African pygmy mice is 3.8 years, what is the power of the test? Give your best approximation. **(5pts)**

For part (a):

$$H_0 : \mu = 3.4, H_1 : \mu > 3.4$$

1 point for hypotheses.

$$T = \frac{\bar{X} - 3.4}{SE(\bar{X})} = \frac{\bar{X} - 3.4}{\sqrt{S_x^2/n}} \sim t_{n-1} \text{ under } H_0 \text{ here } n = 9$$

We need the random sampling assumption. 2 points for statistic and 1 point for t_8 distribution. 1 point for assumption.

For part (b)i:

We reject if $t > t_{95\%,8}$ where $\Pr(T > t_{95\%,8}) = 5\%$, i.e., $t_{95\%,8} = 1.86$. 1 point for rejection rule. 1 point for looking up the right number. Given the realisation, she will obtain $t = \frac{3.7-3.4}{\sqrt{1/9^2}} = 2.7$. 1 point for computed value. Since 2.7 exceeds the critical value, she will reject H_0 at the 5% significance level. 1 point for test result. She found evidence in favour of her claim that African pygmy mice live longer on average. 1 point for interpretation.

For part (b)ii. The p -value gives the lowest level of significance at which you could reject the null.

$$\Pr(T > 2.7) \simeq 1\%$$

(Incorrectly using the normal tables would yield 0.4%)

2 points: demonstrate general understanding of the meaning of p -value

2 points: formal statement on how you obtain the p -value to answer this question

1 point: demonstrate competent use of statistical table – note that any answer in the range 1 – 2.5% should be regarded as correct.

For part (b)iii:

Here we need to obtain the probability of rejecting the null when 3.8 is the true mean:

$$\begin{aligned} & \Pr\left(\frac{\bar{X} - 3.4}{SE(\bar{X})} > 1.86 \mid \mu = 3.6\right) \\ &= \Pr\left(\frac{\bar{X} - 3.8}{SE(\bar{X})} > 1.86 - \frac{0.4}{SE(\bar{X})}\right) \\ &= \Pr\left(T_9 > 1.86 - \frac{0.4}{\sqrt{1/9^2}}\right) = \Pr(T_8 > -1.74) = 0.925 \end{aligned}$$

2 points: proper statement on how to obtain the power

2 points: proper re-standardisation to account for alternative

1 point: proper use of statistical table – note that any answer in the range 90 – 95% should be regarded as correct.

7. Total: 15 points (18 minutes)

- (a) Suppose you wish to choose a probability distribution as an approximating model for a real-world variable. Briefly describe what characteristics of the real-world variable you would take into account when making your choice. **(3pts)**
- (b) Let X and Y represent two continuous random variables, and let x denote an arbitrary realised value of X .
 - i. Explain, in your own words, the difference between $E[Y|X]$ and $E[Y|X = x]$. Do these terms represent random variables or fixed parameters? **(4pts)**
 - ii. Suppose that $E(Y|X = x) = \beta_0 + \beta_1 x$, where the β 's represent fixed parameters. State the law of iterated expectations (LIE). Using the LIE, or otherwise, prove that

$$\text{Corr}(X, Y) = \beta_1 \sqrt{\frac{\text{Var}(X)}{\text{Var}(Y)}}$$

and comment on this finding. **(8pts)**

Suggested answers for part (a) include whether the real-world variable is discrete or continuous, whether the empirical frequencies exhibit skewness or symmetry, and what values the support takes (e.g. positive/negative). 3 points for these or any other reasonable answers - e.g. is it a univariate or vector-valued random variable?

The answer for part (b)i is that the first term is a random variable **1 point**. since X is a random variable. **1 point**. The second term is a fixed parameter **1 point**. governing the distribution of Y conditional on random variable X having realisation x . **1 point**. For part (b)ii, (**2 points for accurate statement of LIE, using whatever notation they wish.**) we first have

$$E(Y) = E_x[E(Y | X = x)] = E_x[\beta_0 + \beta_1 x] = \beta_0 + \beta_1 E(X)$$

And, since

$$\begin{aligned} E(XY) &= E_x[E(XY | X = x)] = E_x[xE(Y | X = x)] \\ &= E_x[x(\beta_0 + \beta_1 x)] = \beta_0 E(X) + \beta_1 E(X^2) \end{aligned}$$

we have

$$\text{cov}(X, Y) = \beta_0 E(X) + \beta_1 E(X^2) - E(X)[\beta_0 + \beta_1 E(X)] = \beta_1 V(X)$$

Thus,

$$\text{corr}(X, Y) = \frac{\beta_1 V(X)}{\sqrt{V(X)V(Y)}} = \beta_1 \sqrt{\frac{V(X)}{V(Y)}}$$

1 point each for $E(Y)$, $E(XY)$, $Cov(X, Y)$, and $Corr(X, Y)$. Thus, $\text{corr}(X, Y) = k\beta_1$, where $k = \sqrt{V(X)/V(Y)} > 0$. In particular, when $\beta_1 = 0$ indicating no linear relationship between X and Y in the sense that $E(Y | X = x) = \beta_0$ does not depend on x , then $\text{corr}(X, Y) = 0$. When $\beta_1 < 0$, then $\text{corr}(X, Y) < 0$. And, when $\beta_1 > 0$, then $\text{corr}(X, Y) > 0$. In general, $\text{corr}(X, Y)$ is reflecting the strength of the linear, or straight-line, relationship between X and Y . **2 points for a sensible comment as above or one that explains the link between correlation and regression coefficients.**

TABLE FD.1 Observations on Income and Education, 20 Observations

I = Observation,
 Y = Income.

Source: Data are artificial.

APPENDIX G



STATISTICAL TABLES

TABLE G.1 Cumulative Normal Distribution. Table Entry Is $\Phi(z) = \text{Prob}[Z \leq z]$

[illegible]

TABLE G.2 Percentiles of the Student's t Distribution. Table Entry Is x Such that $\text{Prob}[t_n \leq x] = P$

n	.750	.900	.950	.975	.990	.995
1	1.000	3.078	6.314	12.706	31.821	63.657
2	.816	1.886	2.920	4.303	6.965	9.925
3	.765	1.638	2.353	3.182	4.541	5.841
4	.741	1.533	2.132	2.776	3.747	4.604
5	.727	1.476	2.015	2.571	3.365	4.032
6	.718	1.440	1.943	2.447	3.143	3.707
7	.711	1.415	1.895	2.365	2.998	3.499
8	.706	1.397	1.860	2.306	2.896	3.355
9	.703	1.383	1.833	2.262	2.821	3.250
10	.700	1.372	1.812	2.228	2.764	3.169
11	.697	1.363	1.796	2.201	2.718	3.106
12	.695	1.356	1.782	2.179	2.681	3.055
13	.694	1.350	1.771	2.160	2.650	3.012
14	.692	1.345	1.761	2.145	2.624	2.977
15	.691	1.341	1.753	2.131	2.602	2.947
16	.690	1.337	1.746	2.120	2.583	2.921
17	.689	1.333	1.740	2.110	2.567	2.898
18	.688	1.330	1.734	2.101	2.552	2.878
19	.688	1.328	1.729	2.093	2.539	2.861
20	.687	1.325	1.725	2.086	2.528	2.845
21	.686	1.323	1.721	2.080	2.518	2.831
22	.686	1.321	1.717	2.074	2.508	2.819
23	.685	1.319	1.714	2.069	2.500	2.807
24	.685	1.318	1.711	2.064	2.492	2.797
25	.684	1.316	1.708	2.060	2.485	2.787
26	.684	1.315	1.706	2.056	2.479	2.779
27	.684	1.314	1.703	2.052	2.473	2.771
28	.683	1.313	1.701	2.048	2.467	2.763
29	.683	1.311	1.699	2.045	2.462	2.756
30	.683	1.310	1.697	2.042	2.457	2.750
35	.682	1.306	1.690	2.030	2.438	2.724
40	.681	1.303	1.684	2.021	2.423	2.704
45	.680	1.301	1.679	2.014	2.412	2.690
50	.679	1.299	1.676	2.009	2.403	2.678
60	.679	1.296	1.671	2.000	2.390	2.660
70	.678	1.294	1.667	1.994	2.381	2.648
80	.678	1.292	1.664	1.990	2.374	2.639
90	.677	1.291	1.662	1.987	2.368	2.632
100	.677	1.290	1.660	1.984	2.364	2.626
∞	.674	1.282	1.645	1.960	2.326	2.576